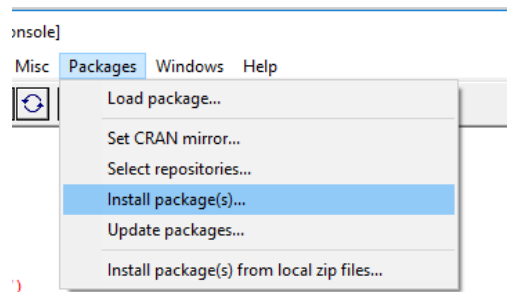


Running BUGS in R**Install OpenBUGS**

Download the zip file from the site

<https://www.mrc-bsu.cam.ac.uk/wp-content/uploads/2014/07/OpenBUGS323setup.zip>

In R, invoke package “ROpenBUGS”

```
> library(R2OpenBUGS)
```

Write the model as a function as save it in a text file in a temporary folder

For example, let's solve one of our homework problems: "Service times of a queuing system follow Exponential distribution with an unknown parameter ϑ . Assume a $\text{Gamma}(\alpha, \lambda)$ prior distribution with $\alpha=4$ and $\lambda=2$. If a sample $X=(0.7, 0.2, 2.5, 3.7, 0.9)$ is observed, find the posterior mean and variance of ϑ .

```
> model <- function() {
  for (i in 1:5){ X[i] ~ dexp(theta) }
  theta ~ dgamma(4,2)
}
> model.file <- file.path(tempdir(), "model.txt")
> write.model(model, model.file)
```

Enter the data, parameters, and initial values

```
> data <- list(X=c(0.7, 0.2, 2.5, 3.7, 0.9))
> params <- list("theta")
> inits <- function(){ list(theta=1) }
```

Run the MCMC, specifying the desired number of iterations

```
> result <- bugs(data, inits, params, model.file, n.iter=1000)
> result
```

```
Inference for Bugs model at
"C:\Users\baron\AppData\Local\Temp\Rtmp4ioVCY\model.txt",
Current: 3 chains, each with 1000 iterations (first 500 discarded)
Cumulative: n.sims = 1500 iterations saved
      mean   sd 2.5%   25%   50%   75%  97.5% Rhat n.eff
theta    0.9 0.3  0.4   0.7   0.9   1.1   1.5    1  1100
deviance 16.0 1.6 14.7 14.9 15.4 16.6  20.3    1  1500
```

For each parameter, n.eff is a crude measure of effective sample size, and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

```
DIC info (using the rule, pD = Dbar-Dhat)
pD = 0.5 and DIC = 16.5
DIC is an estimate of expected predictive error (lower deviance is better).
```

Result: posterior mean = 0.9, posterior variance = 0.09.

Detailed instructions on <http://www.r-tutor.com/bayesian-statistics/openbugs>